LAXMAN MADASU

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in **Generative AI, LLM-based systems, and production-grade AI applications**. Experienced in architecting and deploying **RAG pipelines, AI agent systems, and multi-tenant AI platforms**, with hands-on expertise across LLM orchestration, **structured JSON generation, OCR, and document intelligence**. Built and shipped AI solutions spanning **AI-powered website generation, conversational agents, and financial document processing**, with a strong focus on scalable, provider-agnostic architectures, system reliability, and production readiness.

TECHNICAL SKILLS

Programming & Data Science: Python, NumPy, Pandas, Matplotlib, Seaborn, NLTK, Scikit-learn, OpenCV.

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Data Preprocessing, Hyperparameter Tuning, Model Deployment.

Deep Learning: ANNs, CNNs, RNNs, Data Augmentation, Regularization Techniques, TensorFlow.

Generative AI: LLMs, RAG (dense + BM25 hybrid retrieval), LangChain, AI Agents, Multi-Agent Pipelines, Prompt Engineering, Function/Tool Calling, Structured JSON Generation, OpenAI, Gemini, Anthropic, Groq, DeepSeek, Ollama.

Backend & APIs: FastAPI, REST API Design, Async/Await, SSE Streaming.

Databases & Retrieval: MySQL, MongoDB (Motor), ChromaDB, Vector Search.

Document Intelligence: OCR, PaddleOCR, Tesseract, PyMuPDF, pdfplumber.

Cloud & Application: Streamlit, AWS (EC2, S3, Lambda, SQS), Docker, Docker Compose, Git, GitHub.

PROFESSIONAL EXPERIENCE

Geeth LLC | Waukesha, WI, United States | Remote

AI/ML Engineer | Dec 2024 – Present

Maya - AI Website Generation Platform

Tech Stack: Python · FastAPI · LLMs · RAG · Streamlit · JSON · MongoDB · ChromaDB

- **Engineered** a FastAPI + MongoDB LLM platform converting **natural-language prompts into schema-valid UI page trees for websites**, portfolios, and reusable components.
- **Architected** a multi-stage RAG generation pipeline (**planner → grounded generator → validator → self-repair → quality scorer**) that lifted structure validity from 33% to 100% and component correctness from 98.8% to 100% across a 12-page evaluation set.
- **Implemented hybrid RAG retrieval** (dense embeddings + BM25, fused via Reciprocal Rank Fusion) over a **154-component knowledge base**, eliminating hallucinated component types from generated output.
- **Built** a provider-agnostic LLM layer spanning **7 providers** (Gemini, Groq, OpenAI, Anthropic, Cohere, DeepSeek, Ollama) with cross-provider fallback and per-stage token/cost accounting.
- **Delivered** SSE streaming generation and session version history (undo/revert), validated by a **300-prompt** benchmark corpus and 265 unit tests.

GeethAI - AI Application & Agent Platform

Tech Stack: FastAPI · Python · MongoDB · Streamlit · LLMs · AI Agents · Tool Calling · Multi-Tenant Architecture

- **Developed** a multi-tenant LLM application platform enabling non-engineers to **build and publish AI applications** across three execution modes: **conversational chat, tool-calling agents, and deterministic flows.**
- **Designed** a provider-agnostic LLM abstraction across **7 providers** (OpenAI, Anthropic, Gemini, Groq, Cohere, DeepSeek, Ollama), letting each application swap models purely through configuration.
- **Established** production reliability primitives from scratch — circuit breakers, MongoDB-backed distributed locking, idempotency gating, dead-letter queues, and retry/timeout policies — validated by a **190-test suite.**
- **Enforced strict multi-tenant data** isolation via a database-proxy layer injecting tenant context into every query, combined with deny-by-default tool scoping to block cross-tenant access.
- **Diagnosed** and resolved a critical multi-turn agent reliability issue (tool-call success stuck at 57%) by fixing cross-turn context persistence, restoring reliable multi-step tool execution.

AI Smart Scanner - Finance Application

Tech Stack: Python · FastAPI · Streamlit · MongoDB · OCR · LLMs · LangChain · Docker

- **Built a document-intelligence ETL platform** extracting structured data from **receipts, W-2s, Form 1099 variants, bank statements, and invoices.**
- **Engineered a hybrid OCR pipeline** (PaddleOCR, Tesseract, PyMuPDF, pdfplumber) with automatic engine selection and coordinate-level field mapping, **extracting totals, vendors, and line items without LLM calls**.
- **Integrated a pluggable multi-provider LLM interface** spanning 5 providers (OpenAI, Gemini, Groq, Ollama, DeepSeek) for non-standard document layouts, defaulting to a local-first, privacy-preserving mode.
- **Developed a self-learning transaction classification engine combining rule-based logic** with weighted fuzzy scoring and a feedback loop that promotes high-frequency user corrections into the live rule set.
- **Designed the Streamlit Smart Scanner frontend**, enabling users to **review, correct, and persist extracted financial records into MongoDB.**

EDUCATION

- **Master of Computer Applications (MCA)**
  Jawaharlal Nehru Technological University Hyderabad (JNTUH) | 2024 – 2026
- **Bachelor of Science (Data Science)**
  Satavahana University, Karimnagar | 2020 – 2023

CERTIFICATIONS & TRAINING

- Machine Learning with Python - IBM Developer Skills Network
- Certificate of Course Completion in Data Science - Innomatics Research Labs
- Exploratory Data Analysis Module Completion Certificate - Innomatics Research Labs